

More practice for Quiz 1

IEORE4004, Fall 2025

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1. Your company sells one model of phones and owns one warehouse and one shop. In days 1,2,3, the shop needs to satisfy a given demand $d_1, d_2, d_3 > 0$ of phones, respectively. For $j \in \{1, 2, 3\}$, phones can be shipped from the warehouse to the shop in day j at a given per-unit cost $c_j > 0$. For $j \in \{1, 2\}$, phones already at the shop can be stored between day j and $j + 1$ at a given per-unit cost $q_j > 0$. Assume that the warehouse has enough capacity as to satisfy the demands of the shop.

- (a) Formulate the problem of shipping the phones from the warehouse to the shop in the different days as to satisfy the demand at minimum cost as linear program
- (b) Now suppose that the company also sells a second model of phones. The second model has a demand of d'_1, d'_2, d'_3 in days 1,2, and 3 respectively, and its per-unit shipping cost is c'_1, c'_2, c'_3 in days 1, 2, 3, respectively. The per-unit storing cost of the two types of phone coincide, and there is an upper bound of u_1 on the total number of phones that can be stored between day 1 and day 2, and of u_2 on the total number of phones that can be stored between day 2 and day 3. Given a LP formulation for the problem of satisfying the demand at minimum cost.
- (c) Consider the modification of the problem from the previous bullet point, where the storing constraints are changed as follows:
 - Each phone of the first model takes 1 unit of space, each phone of the second model takes 2 units of space, and we can store phones for w_1 units of space between day 1 and day 2, and w_2 units of space between day 2 and day 3;
 - Each phone of the first model weight 6 oz., each phone of the second model weights 8 oz., and we can store phones for w'_1 oz. between day 1 and day 2, and for w'_2 oz. between day 2 and day 3.

2. Consider the following linear program:

$$\begin{aligned} \max \quad & 4x_1 - 3x_2 + 2x_3 \\ \text{s.t.} \quad & 2x_1 + x_2 - x_3 \geq 20 \\ & -x_1 + 4x_2 + 3x_3 = 25 \\ & 3x_1 - 2x_2 + x_3 \leq 10 \\ & x_1 \geq 0, \quad x_2 \text{ free}, \quad x_3 \leq 0. \end{aligned}$$

- (a) Write the dual of this LP.
- (b) Observe that $x^* = (x_1, x_2, x_3) = (9, 8.5, 0)$ is feasible solution for the problem. Can its dual be infeasible, unbounded, have an optimal solution?
- (c) State the complementary slackness conditions and use them to decide if x^* is optimal for the primal.

3. Consider the following two-players zero-sum game, a modification to the problem seen in Assignment 3. There are two battlefields, with values $v_1 = 3$ and $v_2 = 4$ respectively. Player 1 has 3 units of resources, while player 2 has 4 units of resources. On each battlefield j ($j = 1, 2$),

player 1 wins v_j if they allocate on that battlefield at least as many resources as player 2 (so “a draw” is actually a win for player 1); and loses v_j otherwise. Symmetrically, player 2 wins v_j if they allocate on that battlefield strictly more resources than player 1; and it loses v_j otherwise. Describe the payoff matrix and write explicitly the Linear Programs for computing the optimal strategies. Compute the optimal strategy.

Solutions.

1.

a) Define the following decision variables:

x_j : quantity shipped from the warehouse to the shop on day j , for $j = 1, 2, 3$

s_j : inventory kept from day j to $j+1$ for $j = 1, 2$.

The LP formulation is the following.

$$\begin{aligned} \min \quad & \sum_{j=1}^3 c_j x_j + q_1 s_1 + q_2 s_2 \\ \text{s.t.} \quad & x_1 = d_1 + s_1, \\ & x_2 + s_1 = d_2 + s_2, \\ & x_3 + s_2 = d_3, \\ & x, s \geq 0 \end{aligned}$$

(b) Introduce the additional variables

y_j : shipments of the second model on day j , for $j = 1, 2, 3$

z_j : inventory of the second model kept from day j to $j+1$ for $j = 1, 2$.

The new LP formulation is:

$$\begin{aligned} \min \quad & \sum_{j=1}^3 (c_j x_j + c'_j y_j) + q_1 (s_1 + z_1) + q_2 (s_2 + z_2) \\ \text{s.t.} \quad & x_1 = d_1 + s_1, \\ & x_2 + s_1 = d_2 + s_2, \\ & x_3 + s_2 = d_3, \\ & y_1 = d'_1 + z_1, \\ & y_2 + z_1 = d'_2 + z_2, \\ & y_3 + s_{z2} = d'_3, \\ & s_1 + z_1 \leq u_1, \\ & s_2 + z_2 \leq u_2, \\ & x, y, s, z \geq 0. \end{aligned}$$

(c). Replace the inventory bounds

$$\begin{aligned} s_1 + z_1 &\leq u_1, \\ s_2 + z_2 &\leq u_2, \end{aligned}$$

by the constraints

$$\begin{aligned} s_j + 2z_j &\leq w_j \text{ for } j = 1, 2 \\ 6s_j + 8z_j &\leq w'_j \text{ for } j = 1, 2 \end{aligned}$$

2. (a) The dual is

$$\begin{aligned} \min \quad & 20y_1 + 25y_2 + 10y_3 \\ \text{s.t.} \quad & 2y_1 - y_2 + 3y_3 \geq 4, \\ & y_1 + 4y_2 - 2y_3 = -3, \\ & -y_1 + 3y_2 + y_3 \leq 2, \\ & y_1 \leq 0, \quad y_2 \text{ free}, \quad y_3 \geq 0. \end{aligned}$$

(b). Since the primal is feasible, the dual is either infeasible (if the primal is unbounded) or has an optimal solution (if the primal has an optimal solution).

(c). For a pair of solutions x and y for the primal and the dual,

- Either $x_1 = 0$, or $2y_1 - y_2 + 3y_3 = 4$;
- Either $x_3 = 0$, or $-y_1 + 3y_2 + y_3 = 2$;
- Either $y_1 = 0$, or $2x_1 + x_2 - x_3 = 20$;
- Either $y_3 = 0$ or $3x_1 - 2x_2 + x_3 = 10$.

Since $x_1 > 0$, we deduce

$$2y_1 - y_2 + 3y_3 = 4.$$

Since $2x_1 + x_2 - x_3 > 20$, we deduce

$$y_1 = 0.$$

Together with $y_1 + 4y_2 - 2y_3 = -3$, we obtain a system of 3 equations in the 3 variables y_1, y_2, y_3 , whose unique solution is $y^* = (0, -1/10, 13/10)$. As this is feasible for the dual, we conclude that x^* is optimal for the primal and y^* is optimal for the dual.

3. We let player 1 be the column player (cp) and player 2 be the row player (rp). As in Assignment 3, Problem 1, the strategies are:

$$\text{cp strategies (columns): } C_1 = (0, 3), C_2 = (1, 2), C_3 = (2, 1), C_4 = (3, 0).$$

Similarly, rp's strategies are all the ordered pairs of 2 numbers that sum to 4— that is, rp's budget. These are,

$$\text{rp strategies (rows): } R_1 = (0, 4), R_2 = (1, 3), R_3 = (2, 2), R_4 = (3, 1), R_5 = (4, 0).$$

The payoff matrix from the point of view of the cp is (we highlight in bold the difference with the solution to Assignment 3, Problem 1):

$$A = \begin{pmatrix} -\mathbf{1} & -1 & -1 & -1 \\ \mathbf{1} & -\mathbf{1} & -1 & -1 \\ 1 & \mathbf{1} & -\mathbf{1} & -1 \\ 1 & 1 & \mathbf{1} & -\mathbf{1} \\ 1 & 1 & 1 & \mathbf{1} \end{pmatrix}.$$

LP for cp. Following the same template as in the notes, the optimal mixed strategy for the cp (probabilities on the *columns*) is given by the solution of

$$\begin{aligned}
& \max \quad v \\
& \text{s.t.} \quad -x_1 - x_2 - x_3 - x_4 \geq v && (\text{row } R_1) \\
& \quad \quad x_1 - x_2 - x_3 - x_4 \geq v && (\text{row } R_2) \\
& \quad \quad x_1 + x_2 - x_3 - x_4 \geq v && (\text{row } R_3) \\
& \quad \quad x_1 + x_2 + x_3 - x_4 \geq v && (\text{row } R_4) \\
& \quad \quad x_1 + x_2 + x_3 + x_4 \geq v && (\text{row } R_5) \\
& \quad \quad x_1 + x_2 + x_3 + x_4 = 1, \\
& \quad \quad x_1, x_2, x_3, x_4 \geq 0.
\end{aligned} \tag{CP}$$

LP for rp. Symmetrically, the optimal mixed strategy for the rp (probabilities on the *rows*) is obtained from

$$\begin{aligned}
& \min \quad u \\
& \text{s.t.} \quad -y_1 + y_2 + y_3 + y_4 + y_5 \leq u && (\text{column } C_1) \\
& \quad \quad -y_1 - y_2 + y_3 + y_4 + y_5 \leq u && (\text{column } C_2) \\
& \quad \quad -y_1 - y_2 - y_3 + y_4 + y_5 \leq u && (\text{column } C_3) \\
& \quad \quad -y_1 - y_2 - y_3 - y_4 + y_5 \leq u && (\text{column } C_4) \\
& \quad \quad y_1 + y_2 + y_3 + y_4 + y_5 = 1, \\
& \quad \quad y_1, y_2, y_3, y_4, y_5 \geq 0.
\end{aligned} \tag{RP}$$

To compute the optimal strategy, note that rp can always get a profit of 1 if they play (0, 4) (move corresponding to the first row)– and that’s the maximum profit they can get no matter what they play. So the row player will play that strategy with probability 1. Any strategy of the column player is an optimal strategy (or, said otherwise, it does not matter which strategy the column player plays).